

PULLANI MADHU

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PROFESSIONAL SUMMARY

Generative AI Developer with 2 year of experience at Tata Consultancy Services, building production-grade AI solutions across enterprise platforms. Hands-on expertise in LangGraph, RAG architectures, multi-agent systems, and LLM-powered automation. Contributed to high-impact projects including SAP documentation generation, vendor self-service chatbots, and intelligent data analytics. Adept at integrating AI solutions with enterprise systems like SAP S/4HANA, ServiceNow, and Brainware.

EXPERIENCE

Generative AI Developer

Tata Consultancy Services (TCS), India

June 2024 – Present

- Contributed to the **iR2D Multi-LLM Orchestration Platform**, working on the **Functional Spec Agent** — owning the complete agent pipeline including RAG design, agent orchestration, prompt engineering, and LangGraph workflow integration.
- Developed the **P2P Assistant** enabling vendors to check PO status and raise ServiceNow tickets, with integrations across SAP S/4HANA, Brainware, and ServiceNow REST APIs.
- Built the **Data Insights Assistant** — an NLP-driven SQL agent with 90% query-to-SQL translation accuracy, delivering graphical insights through a ReactJS dashboard with dynamic chart visualizations.
- Implemented RAG pipelines and multi-agent orchestration workflows to automate SAP documentation generation including BRDs, Functional Specs, Test Cases, Technical Specs, and ABAP code.
- Contributed to a **50–60% reduction** in documentation effort and **40% reduction** in code development/review cycles across the software delivery lifecycle.

KEY PROJECTS

iR2D – Multi-LLM Orchestration Platform

LangGraph, Multi-Agent Systems, RAG, SAP Integration, LangChain, OpenAI

2024 – Present

- Owned the **Functional Spec Agent** end-to-end: designed the RAG pipeline, built agent orchestration logic, and integrated with LangGraph for automated SAP Functional Specification document generation.
- Worked across a platform orchestrating multiple LLMs for user story processing, design documents, code generation, code review, and test management.
- Delivered measurable effort reductions: **50–60%** in documentation, **40%** in code development/review, and **30%** in training content generation.
- Integrated multi-cloud compatible architecture with ERP (SAP S/4HANA) connectivity and retrieval augmentation.

P2P Assistant – Vendor Self-Service Chatbot

LangChain, SAP S/4HANA, Brainware, ServiceNow, Python, FastAPI

2025 Feb – 2025 Aug

- Built a conversational AI assistant allowing vendors to autonomously check purchase order status and raise ServiceNow tickets when issues arise.
- Developed integrations with SAP S/4HANA REST APIs, Brainware document processing, and ServiceNow ticketing workflows.
- Implemented secure role-based access control and API authentication for enterprise-grade ERP data retrieval.

Data Insights – Intelligent Analytics Platform

LangGraph, SQL Agents, NLP, ReactJS, Python, Generative AI

2024 Oct– 2025 Jan

- Developed an NLP-powered platform that translates natural language queries into SQL with **90% accuracy** using agentic prompt engineering workflows.
- Built a dynamic visualization engine generating bar, pie, and line charts with automated narrative insights for business decision-making.
- Delivered an end-to-end pipeline: data ingestion → NLP query → SQL execution → graphical output rendered in a ReactJS dashboard.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, SQL, C

Generative AI & LLMs: LangChain, LangGraph, CrewAI, OpenAI GPT, Claude, Gemini, LLaMA, Ollama, Hugging Face Transformers

AI Techniques: RAG, Prompt Engineering, Multi-Agent Systems, Agentic Workflows, NeMo Guardrails, Cache Augmented Generation

Machine Learning: Scikit-learn, TensorFlow, PyTorch, Keras, Neural Networks, NLP, Computer Vision

Web Development: ReactJS, Flask, Django, FastAPI, Node.js, HTML, CSS, RESTful APIs

Databases: PostgreSQL, MySQL, MongoDB, Neo4j, FAISS, Chroma, Redis

Cloud Platforms: Microsoft Azure, AWS, Google Cloud Platform, Azure AI Services, Azure Machine Learning

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Apache Spark, Feature Engineering

Integrations: SAP S/4HANA REST APIs, ServiceNow, Brainware

Miscellaneous: PowerBI, Cryptography, Problem Solving

EDUCATION

Vardhaman College of Engineering, Hyderabad	Jul 2019 – Jun 2023
Bachelor of Technology in Computer Science and Engineering	GPA: 7.03/10

CERTIFICATIONS

- **Microsoft Certified:** Azure AI Engineer Associate (AI-102) – Designing and Implementing Azure AI Solutions
- **SAP Certified:** Generative AI Developer
- Data Analytics Real-World Projects in Python – Udemy
- Programming in Java – NPTEL
- Artificial Intelligence – Personify-1stop